

Variables Advanced

Learning Objectives

- Create and use variables in Scratch projects
- Understand the difference between local and global variables
- Implement variables for score tracking and game mechanics

Engage (5-7 minutes)

In farming, a farmer tracks how many bags of rice were harvested or how much rain fell. Variables in Scratch work the same way — they store numbers and text that can change as the program runs, just like a farmer’s harvest count increases through the season.

Explore (10-12 minutes)

Students experiment with creating variables: “score,” “timer,” and “lives.” They see how variables appear as watchers on the stage, like a farmer’s ledger showing current crop yield. Different blocks change variable values — set, change by, and show/hide.

Explain (10-12 minutes)

Variables are containers that store data. In Scratch, variables can hold numbers (like score) or text (like player name). Local variables exist only within a sprite, while global variables are accessible everywhere — like village-wide records versus individual farm records.

Elaborate (8-10 minutes)

Students build a simple counter: a sprite clicks to increase a score variable, then resets it. Like a shopkeeper counting daily sales, variables keep track of important numbers that change during program execution.

Evaluate (5-8 minutes)

Can students create variables and understand their purpose? Can they explain when to use local versus global variables? Do they successfully implement a score counter in their project?